Major Project — Problem Statement Shortlist

ADCMS · Application Development Course Management System · ADYPU

Team: Vivek Krishna · Aditya TD · Aman Verma | Project type: **Major Project** | Shortlist: 24 ideas

Reviewed the full pool available under Major Project (368 statements). The list below filters it to **software-development projects a 3-person team can realistically design, build, demo and document** — hardware-heavy SIH challenges were excluded. Sections 1–2 are engineering-heavy picks; Section 3 covers applied, real-world-impact apps (market access, agriculture, livelihood); Section 4 lists further software statements worth knowing. Each **PS #** is the item's position number in the ADCMS "Select a Problem Statement" dropdown (Major Project).

1. TOP PICKS — SHORTLISTED FOR THE TEAM

1 deplo.ai — AI-Assisted Deployment & Cloud Orchestration Platform

PS #343 STUPS44 **MAJOR / DESKTOP APP**

deplo.ai removes that step entirely. The user signs in with GitHub, connects their own Vercel and Render accounts once, picks a repository, and the platform does the rest — it analyses the codebase to detect the stack and build settings, provisions projects on both providers, deploys the backend first, injects the resulting URL into the frontend's environment, deploys the frontend, streams the live build log throughout, and returns two working production URLs. Importantly, deplo.ai is not a hosting company. Applications are deployed into the user's own provider accounts through per-user OAuth, so the developer keeps ownership, control and billing of their infrastructure while deplo.ai supplies only the intelligence and the orchestration.

> **Why it fits a 3-person team:** Most ambitious and most impressive on a resume. Splits cleanly into 3 workstreams: OAuth + provider API layer, the deploy-orchestration engine, and the React dashboard.

2 VoxNovel: Multi-Voice Audiobook Synthesis with a Synchronized Web Reader

PS #348 PSNEW03 **WEB APP / AI + BACKEND**

Most digital novels, web serials and e-books are read aloud by a single-voice text-to-speech engine, which makes dialogue hard to follow. VoxNovel is a multi-voice audiobook platform that turns plain novel text into cast-narrated audio and plays it in a synchronized web reader. The system runs as a pipeline. Pasted text is first cleaned in the browser to remove page clutter. A rule-based NLP stage then uses regular expressions and spaCy dependency parsing to separate narration from dialogue and attribute each line to a character, flagging uncertain lines for manual review. Each character is assigned a voice from a normalized catalog, which the user can override. A background worker, coordinated through PostgreSQL 16 with FOR UPDATE SKIP LOCKED job claiming, synthesizes speech with the Kokoro-ONNX neural model. Unchanged segments are reused from a cache, and timestamps are calculated from cumulative audio sample counts, then encoded to Opus in a single pass to avoid timing drift. A React reader streams synthesis progress through Server-Sent Events, highlights the active passage in time with the audio, and lets listeners jump to any line. The whole system runs from a single Docker Compose setup.

> **Why it fits a 3-person team:** The prompt effectively hands you the architecture (NLP attribution → job queue → TTS → SSE reader), so effort goes into building rather than guessing. Great AI + full-stack split.

3 API Sentinel: API Failure Detection, Breaking-Change Analysis, and MCP-Based Diagnostics.

PS #359 PSNEW24 **BACKEND / DEVOPS TOOLING**

Modern software applications depend on APIs to exchange data and provide essential services. However, API failures, slow responses, and incompatible contract changes can disrupt dependent applications and affect user experience. Developers often need to identify failures, investigate incidents, understand changes between API versions, and determine which integrations may be affected. API Sentinel aims to address these challenges through a unified platform for API monitoring, breaking-change analysis, impact assessment, and AI-assisted diagnostics. The system monitors registered API endpoints and records information such as HTTP status codes, response latency, check outcomes, and incident details. It compares different versions of API specifications to identify potentially breaking changes, including removed fields, newly required fields, and incompatible data-type modifications. The platform also maintains integration dependency information to help identify consumers that may be affected by detected changes. Additionally, API Sentinel uses the Model Context Protocol (MCP) to expose selected backend capabilities through structured diagnostic tools that compatible AI applications can invoke. These tools can help retrieve endpoint health information, inspect incidents, compare API contracts, and investigate potential impacts. By combining monitoring, contract analysis, dependency tracking, and structured diagnostics, API Sentinel aims to help developers detect problems earlier, understand compatibility risks, and make troubleshooting more systematic, reliable, and evidence-based.

> **Why it fits a 3-person team:** Modern, in-demand domain (observability + MCP). Modules are independent: monitoring service, OpenAPI diff engine, dependency graph, AI diagnostics.

4 GridLock: Spatio-Temporal Traffic Mobility Predictor

PS #358 PSNEW22 **ML / DATA SCIENCE**

The GridLock Mobility Predictor is a high-performance machine learning pipeline engineered to forecast urban traffic demand and congestion across dynamic spatio-temporal environments by transforming raw geospatial coordinates, timestamps, weather telemetry, and urban landmarks into actionable intelligence. Utilizing domain-driven feature engineering like geohash clustering and 24-hour cyclical day-over-day lag dynamics, the system enables accurate long-horizon traffic extrapolation up to 13 hours into the future. Its core architecture employs a stacked ensemble of LightGBM, XGBoost, and CatBoost regressors harmonized through an empirical score-geometry blending layer. By enforcing strict chronological validation to eliminate data leakage, GridLock achieved a top-tier 91.45 / 100 R^2 score on the Flipkart GridLock 2.0 competition leaderboard, offering a scalable, production-ready foundation for municipal traffic planning, intelligent fleet dispatching, and automated congestion mitigation in connected smart cities.

> **Why it fits a 3-person team:** Ideal if one teammate wants an ML-heavy portfolio piece: feature engineering, model ensemble, time-series validation. Reproducible from public traffic data.

5 Fragmented Chess Telemetry & Longitudinal Skill Stagnation

PS #351 PSNEW10 **BACKEND / DATABASES**

Modern digital chess platforms evaluate millions of games daily, yet their analytical architectures remain ephemeral and session-isolated. Post-game analysis offers point-in-time feedback on individual moves, but fails to aggregate historical telemetry into actionable diagnostic trends. Consequently, competitive players cannot systematically identify whether an error reflects a recurring tactical blind spot across specific openings, a phase-dependent calculation collapse, or chronic time-decay patterns. Compounding this diagnostic gap is an architectural trade-off in existing platforms: move evaluations are either stored as denormalized, unstructured JSON blobs or wide relational records plagued by NULL values and lock contention. These monolithic structures prevent granular, cross-game indexing and make point-in-time temporal tracking computationally prohibitive. Furthermore, commercial platforms often employ arbitrary early-ply heuristics that mask critical opening blunders from empirical player ratings. The challenge is to design an enterprise-grade longitudinal analytics engine capable of ingesting high-throughput game data into a fully normalized relational schema without incurring database write-amplification bottlenecks. The system must reliably isolate concurrent worker executions, enforce rigorous evaluation mathematics that account for board perspective and forced-mate continuities, and transform aggregate telemetry into targeted, personalized tactical training regimens.

> **Why it fits a 3-person team:** A systems + database design showcase (normalization, worker concurrency, aggregation pipelines). Strong for a backend-focused resume.

6 Run & Conquer: GPS-Based Running and Territory Conquest Platform

PS #356 PSNEW20 **MOBILE + GEOSPATIAL**

Run & Conquer is a gamified fitness platform that transforms outdoor running into an interactive territory-conquest experience. Users can record their runs using GPS, claim geographic territories by completing qualifying routes, strengthen their territories through repeated running activities, and compete with other runners for control of nearby areas. The platform features an interactive map displaying territories and their ownership, user registration and profiles, running history, GPS-based activity tracking, territory claiming and conquest mechanisms, location-based leaderboards, and community competitions. The system validates running activities before updating territory ownership to reduce fraudulent claims and maintain reliable records. The project aims to encourage regular physical activity, exploration, and healthy competition by making running more engaging and rewarding.

> **Why it fits a 3-person team:** Fun and demo-friendly. Splits into a map/geo engine, a run-tracking client, and an anti-fraud/leaderboard backend.

2. STRONG ALTERNATIVES

7 Project Title: DisasterReady – Gamified Disaster Preparedness and Emergency Response Training

PS #357 PSNEW21 **GAME / SIMULATION**

DisasterReady is an interactive disaster management gaming platform designed to prepare people to respond effectively to unexpected real-life emergencies through immersive, game-based simulations. The platform places players in realistic disaster scenarios such as earthquakes, floods, fires, and cyclones, where they must make quick decisions, identify hazards, choose safe evacuation routes, and rescue people. Inspired by multiplayer survival games like BGMI and Free Fire, it incorporates unexpected disaster events, dynamic environments, time-based challenges, and cooperative missions to make learning engaging and practical. The game evaluates players' decisions, demonstrates the consequences of unsafe actions within the simulation, and provides feedback on appropriate emergency responses. Players can repeat missions to strengthen their understanding, improve decision-making, and develop essential survival skills. By combining entertainment with disaster preparedness education, DisasterReady aims to make emergency training accessible, engaging, and repeatable for students, families, employees, and communities, helping them build awareness, confidence, and practical skills to respond more safely and effectively when real-life disasters occur.

> **Why it fits a 3-person team:** Gamified disaster training. Game loop + scenario engine + scoring with a simple UI.

8 TradeWise: An Educational Stock-Market Learning and Paper-Trading Platform

PS #354 PSNEW16 **FINTECH / WEB APP**

Many students and beginners are interested in learning stock-market trading but lack a safe and practical environment to develop their skills. Trading with real money involves financial risk, especially for individuals who have limited knowledge of market behavior, order execution, portfolio management, and risk management. Traditional theoretical learning methods may not provide sufficient hands-on experience, making it difficult for beginners to understand how trading decisions affect portfolio performance. There is a need for an educational platform that allows users to practice trading, analyse market data, and evaluate strategies without risking real money. The platform should provide simulated trading, portfolio tracking, market charts, historical market replay, strategy backtesting, and interactive learning resources in one integrated system. TradeWise aims to address this problem by providing a virtual trading environment where users can place simulated orders, monitor portfolio performance, practice strategies using historical data, and improve their financial knowledge through lessons and quizzes. By combining practical simulation with structured learning, the platform seeks to help beginners build confidence, develop disciplined trading habits, and gain a better understanding of stock-market operations before considering real-world trading.

> **Why it fits a 3-person team:** Paper-trading + backtesting + lessons. Charts, order engine, and backtester are natural splits.

9 CampusFind

PS #355 PSNEW19 **FULL-STACK WEB**

is a web-based Lost-and-Found Management System designed to help college students recover misplaced belongings easily. Students often lose items such as ID cards, wallets, keys, books, and earphones, while information about found items is scattered across messaging groups, college offices, and word of mouth. This makes finding lost belongings difficult and time-consuming. CampusFind provides a centralized platform where users can report lost items and post details of items they have found, including the item name, category, description, date, and location. Users can search, browse, and filter reports to find relevant items. The system also suggests potential matches between lost and found reports based on item descriptions, categories, dates, and locations. Students can review possible matches, contact the person who posted the report, verify ownership, and arrange the return of the item. Users can also update the status of reports when belongings are recovered. The main objective of CampusFind is to simplify the lost-and-found process, improve communication among students, reduce the time spent searching for misplaced belongings, and provide an organized and accessible solution for managing lost and found items within a college campus.

> **Why it fits a 3-person team:** Smallest safe scope on this list. Ship it confidently; great fallback if you want low risk.

10 AeroGrid Intelligence — Micro-Airstrip Spatial Decision Platform

PS #349 PSNEW04 **GEO / FULL-STACK**

AeroGrid Intelligence is a web-based spatial data engineering platform that turns fragmented, offline records of regional micro-airstrips into an interactive map and decision tool. Hundreds of underutilised paved, unpaved, private, municipal and state-owned airstrips lack reliable, comparable data, so operators such as eVTOL air-taxi, STOL, charter, cargo and air-ambulance companies struggle to decide where to base or route aircraft. The platform stores each airstrip's runway specifications, surface, fuel and night-landing capability, together with terrain and obstacle risks, catchment population, nearby hospitals and last-mile ground access times. Spatial queries in PostgreSQL with PostGIS find airstrips within a radius, the nearest options and the links a fleet can fly between them. A later phase adds aircraft factors such as payload, passenger capacity, range and fuel, so the system can report whether a specific aircraft fits a specific airstrip, with reasons. The application uses a React and Leaflet map, a Node.js and Express API on Zoho Catalyst, and a normalised PostgreSQL database on Neon. JWT authentication, role-based access control, structured logging and central error handling support secure, reliable use. Built for planners, operators and aviation departments, AeroGrid supports faster, evidence-based infrastructure and fleet deployment decisions, replacing scattered paperwork with one trusted, queryable source.

> **Why it fits a 3-person team:** The entire stack is specified (React + Leaflet, Node/Express, PostGIS). Spatial queries are the interesting core.

11 Strata — CLI Tool for Git Repository History Analysis

PS #335 STUPS28 **TERMINAL CLI**

The tool must run Git as a subprocess and read arbitrary directories on the local machine, neither of which a browser permits. JSON output allows it to be piped into other tools, and it can run inside an automated pipeline with no human interaction. The intended user is already working in a terminal inside a repository, so a single command is faster than any graphical interface.

> **Why it fits a 3-person team:** Elegant, self-contained tool. Smaller than the others, but very polished as a portfolio piece.

12 Nura: A Personalized Menstrual Health Tracking and Awareness System

PS #352 PSNEW12 **MOBILE / HEALTH**

Nura is a menstrual health companion designed to help users track their periods, record symptoms, understand their cycle patterns, and access relevant health information. The idea behind Nura is that menstrual health is not just about predicting the next period. It's also about helping users recognize patterns in their own records and make more informed decisions about their health. Our system brings together period tracking, symptom logging, cycle insights, and health reports in one place, with a focus on privacy and user awareness.

> **Why it fits a 3-person team:** Privacy-focused health tracker. Clear, responsible scope with real user value.

13 Community-Driven Personalized Mathematics Learning Roadmap Platform

PS #353 PSNEW15 **WEB / COMMUNITY**

Mathematics learning resources are scattered across multiple platforms, making it difficult for learners to discover engaging content and follow a structured learning path based on their interests. I am building a community-driven web application that generates personalized mathematics roadmaps based on each learner's interests, current knowledge, and goals. Whether someone wants to explore Linear Algebra, Probability, Statistics, Calculus, Number Theory, or Discrete Mathematics, the platform will help them discover relevant topics and understand how different mathematical concepts connect. Each roadmap will include a diverse collection of learning resources, such as video lectures, articles, interactive games, visual simulations, thought-provoking problems, and practical applications that make abstract mathematical ideas easier to understand and explore. Rather than following a generic curriculum, learners will be able to choose their areas of interest and follow a learning journey tailored to their preferences and pace. The platform will also encourage community participation by allowing users to contribute resources, share interesting problems, and improve existing roadmaps. The goal is to make mathematics more accessible, interactive, and enjoyable by bringing high-quality resources together in one place and helping learners explore mathematical concepts beyond traditional textbooks and classroom learning.

> **Why it fits a 3-person team:** Community-driven math roadmaps. Content library + recommendation engine + contribution flow.

14 Safar — Android-First Travel Planning App

PS #364 STUPS58 **MOBILE / MAJOR**

Make travel planning understandable, flexible and accessible for people around the world—from major cities to smaller towns and custom destinations. A user should be able to create a trip, maintain its details, and share a plan without being required to use AI.

> **Why it fits a 3-person team:** Android-first travel planner where AI is optional. Manual-planning-first keeps scope in control.

15 BuildX — AI Blueprint-to-Code Desktop Platform

PS #368 STUPS63 **AI / DESKTOP**

BuildX is an AI-powered application development platform that helps users turn a simple app idea into a structured technical blueprint and working project. The existing version of BuildX can analyze an idea and generate important parts of the project such as its architecture, technology stack, database schema, APIs, UI structure and implementation plan. It also provides an AI-assisted coding workspace where generated project files can be viewed, modified, previewed and refined. For the major project, BuildX will be extended into a desktop application, bringing its existing AI planning and coding capabilities into a more complete local development environment. The goal is to create a smoother workflow where users can go from: Idea → Blueprint → Code → Preview → Refinement within a single desktop application.

> **Why it fits a 3-person team:** Idea → blueprint → code workflow. Ambitious; strong if your team leans AI tooling.

3. APPLIED / REAL-WORLD-IMPACT APPS (MARKET ACCESS, AGRICULTURE, LIVELIHOOD)

16 App-Based Solution to identify and solve disease in plants/crops

PS #253 SIH1401 **MOBILE / AI**

Farmers can upload the photographs of disease affected plants/crops and solutions may be provided by experts/scientists linked to the app. Source metadata Year: 2023 Domain / Technology Bucket: Agriculture, FoodTech & Rural Development Category (SW/HW): Software Organisation: Ministry of Micro, Small and Medium Enterprises

> **Why it fits a 3-person team:** This is the "AI crop disease detector" you meant — a Software-category statement under the Ministry of MSME. Farmers upload photos of affected crops; experts/AI return a diagnosis and remedy. Simple to demo, high impact.

17 Farmers Disease Diagnostic/Reporting Portal - Mobile Portal AI Based

PS #76 SIH1673 MOBILE / AI

Background- Agricultural productivity and food security are heavily dependent on the health of crops and livestock. Farmers, especially in remote or resource-limited areas, often face challenges in diagnosing and reporting diseases that affect their livestock and crops. Early diagnosis and timely reporting are crucial for effective disease management and prevention of widespread outbreaks. Description - A mobile portal powered by Artificial Intelligence (AI) for disease diagnosis and reporting can revolutionize how farmers manage animal and plant health. This portal enables farmers to use their smartphones to access diagnostic tools, report symptoms, and receive actionable advice. AI algorithms can analyze reported data to provide accurate diagnoses and suggest appropriate treatments or interventions. This technology can empower farmers with the knowledge and tools to manage diseases more effectively, ensuring the health and productivity of their farms. Integrate AI-based software with existing NDLM to allow farmers to submit images and descriptions of disease signs and symptoms. The AI will generate suspected disease/condition reports with preventive measures and send alerts to veterinarians for appropriate action (AI-based software, Mobile application). Expected Outcomes . Enhanced Disease Diagnosis: . Timely Reporting and Intervention: . Increased Access to Expert Knowledge: . Improved Farm Productivity. Data Collection and Analysis: Cost-Effective DiseaseManagement: Empowerment and Education of Farmers: Integration with Existing Surveillance Systems . Community Engagement and Support: Sustainable Agriculture Practices Source metadata Year: 2024 Domain / Technology Bucket: Agriculture, FoodTech & Rural Development Category (SW/HW): Software Organisation: Ministry of Fisheries, Animal Husbandry and Dairying Department: Department of Animal Husbandry and Dairying

> **Why it fits a 3-person team:** AI diagnostic + reporting portal for crop and livestock disease, aimed at remote farmers. Same idea as above but with a reporting/escalation flow.

18 Development of AI-ML based models for predicting prices of agri-horticultural commodities such as pulses and vegetable (onion, potato, onion)

PS #62 SIH1647 ML / WEB APP

The Department of Consumer Affairs monitors the daily prices of 22 essential food commodities through 550 price reporting centres across the country. The Department also maintains buffer stock of pulses, viz., gram, tur, urad, moon and masur, and onion for strategic market interventions to stabilize the volatility in prices. Decisions for market interventions such as release of stocks from the buffer are taken on the basis of the price trends and outlook. At present, the analyses of prices are based on the seasonality, historical and emerging trends, market intelligence inputs, crop sowing and production estimates. ARIMA based economic models have also been used to examine and forecast prices of pulses. Source metadata Year: 2024 Domain / Technology Bucket: Agriculture, FoodTech & Rural Development Category (SW/HW): Software Organisation: Ministry of Consumer Affairs, Food and Public Distribution Department: Department of Consumer Affairs

> **Why it fits a 3-person team:** AI-ML model to predict prices of agri-horticultural commodities. Data-driven market insight for farmers, traders and policy.

19 Application Development for monitoring of wool from Farm to fabric

PS #183 SIH1309 MOBILE APP

An App-based solution for the wool sector in India which can have the following features is required: 1. Wool Market Information: Provides real-time market information on wool prices, trends, and news. 2. Wool Tracking: Allows users to track the production and transport of their wool from farm to market. 3. Quality Assurance: Provides a platform for wool producers to ensure the quality of their wool and access to wool grading services. 4. Wool Storage and Warehousing: Helps farmers store and manage their wool inventory. 5. Wool Processing: Helps farmers access wool processing services such as shearing, sorting and dyeing 6. Wool Trading Platform: Allows wool producers to buy and sell wool directly from other farmers or buyers. 7. Online Wool Marketplace: Provides a platform for wool producers to sell their wool directly to buyers 8. Wool Education and Training: Provides a list of wool producers/artisans region-wise, and state-wise data to provide educational resources and training to help them improve their production, quality, design and marketing skills. Source metadata Year: 2023 Domain / Technology Bucket: Transportation & Logistics Category (SW/HW): Software Organisation: Ministry of Textiles

> **Why it fits a 3-person team:** Farm-to-fabric wool tracking app with real-time market prices, quality assurance and buyer access — market access for small producers.

20 Creation of dedicated real time 'Procurement Data Portal'as well as 'Vendor Payment Data portal'to capture all related data in respect of NEEPCO,S Procurements and Payments' including Procurement from MSE Sector and procurement in GeM Portal

PS #319 SIH1508 WEB PORTAL / FULL-STACK

This challenge involves developing a comprehensive online portal for North Eastern Electric Power Corporation (NEEPCO) to capture and manage all data related to their procurement processes and vendor payments. The portal must have the capability to record procurement activities, particularly those involving Micro and Small Enterprises (MSE) and purchases made through the Government e-Marketplace (GeM) portal. The solution should enable efficient data entry, storage, retrieval, and analysis of procurement details, payment records, and vendor information. It should also ensure data security and compliance with relevant regulations. By addressing this challenge, the aim is to enhance transparency, streamline procurement operations, promote MSE participation, and facilitate effective financial management for NEEPCO. Source metadata Year: 2023 Domain / Technology Bucket: Miscellaneous Category (SW/HW): Software Organisation: Ministry of Power

Why it fits a 3-person team: Closest match to your “market access for MSEs” idea: a procurement + vendor-payment portal covering the MSE sector and GeM procurement.

21 Price comparison of GeM products with other e-marketplaces

PS #226 SIH1360 WEB / DATA

The problem at hand involves developing a cost or price comparison solution specifically tailored for comparing the prices of products available on GeM (Government e-Marketplace) with other e-marketplaces or e-commerce platforms. GeM is a dedicated online platform in India that facilitates the procurement of goods and services by various government departments, organizations, and public sector undertakings. Comparing the prices of products listed on GeM with those on other platforms is crucial for ensuring fair and competitive pricing. The cost or price comparison solution aims to provide users with a convenient way to compare the prices of products available on GeM with other popular e-marketplaces or e-commerce platforms. By leveraging data scraping techniques, APIs, and data analytics, the solution will gather and analyze pricing information from multiple sources, allowing users to make informed decisions based on the best available options. Source metadata Year: 2023 Domain / Technology Bucket: Smart Automation Category (SW/HW): Software Organisation: Ministry of Commerce and Industries

Why it fits a 3-person team: Price comparison of GeM products against other e-marketplaces — aggregation, scraping and analytics.

22 India Post A Bridge for Indian Diaspora to access things Indian " Building a community of Indian Diaspora for meeting their needs of Indian Products (traditional/ ethnic / handicrafts) through India Post by connecting PIOs with local sellers / MSME / Artisans"

PS #136 SIH1757 WEB / MARKETPLACE

Background: Department of Posts (DoP) covers around 219 countries where a large diaspora of India in different economic capacities exists. PIOs still take/use Indian Products such as Puja Samagri/Traditional Attires/ Handicrafts. Most of them are dependent on e-commerce platform. Through this they cannot be assured of the quality of the products they get and transportation cost may also be very high. Secondly local sellers/small handicrafts artisans who are the actual manufacturers are not able to ship high quality products because of limited resources and information at their disposal. Hence there is a need to connect the Indian Community abroad with the local artisans. Description: In order to overcome this issue an application is required through which one can go to a particular local seller as per their preference and should be able to onboard the seller as well. The following data points may be required to be integrated in the said Application. 1. Global Indian Diaspora Data on the basis of their needs namely – Income/ Financial data, Consumption pattern and buying behavior pattern. 2. There should be Buyer and Seller Registration so that Quality Assurance and standard Transit Cost can be ensured. 3. Product and customer segments identification of target population, this includes utilizing Artificial Intelligence (AI) and Machine Learning (ML) so that specific needs can be assessed like Puja Samagri of Kumartuli, Handicrafts of Santiniketan, Moya of Joynagar (famous sweets), Terracotta Arts of Bishnupur, etc. 4. Special preference should be given on keeping in the mind of Seasonal needs, Traditional Attires for Wedding ceremony, Festival needs as this varies geographically depending upon Special Socio Economic Culture. 5. A range of products which are prepared (by MSME and others) in the different parts of India , which can match the buying needs of global diaspora. So that a global link through Postal Export may be created towards fulfillment of needs. Therefore, it is about bringing the Diaspora close to India and remains emotionally connected to their roots. Source metadata Year: 2024 Domain / Technology Bucket: Transportation & Logistics Category (SW/HW): Software Organisation: Ministry of Communication Department: Department of Posts

Why it fits a 3-person team: Marketplace connecting the Indian diaspora with local sellers, MSMEs and artisans (India Post as the bridge).

23 Sustainable Fertilizer Usage Optimizer for Higher Yield

PS #57 SIH1639 MOBILE / DATA

Background: Excessive and improper use of fertilizers leads to soil degradation and reduced agricultural productivity, negatively impacting farmers' income. Description: Create a data-driven solution that recommends optimal fertilizer types and quantities based on soil health, crop type, and weather patterns, ensuring sustainable agricultural practices. Expected Solution: An application that analyzes soil data and provides tailored fertilizer recommendations, promoting sustainable farming while enhancing crop yield and farmer income. Source metadata Year: 2024 Domain / Technology Bucket: Agriculture, FoodTech & Rural Development Category (SW/HW): Software Organisation: Ministry of Agriculture and Farmers Welfare Department: Indian Council of Agricultural Research (ICAR)

> **Why it fits a 3-person team:** Fertilizer-usage optimizer app that recommends optimal fertilizer type/quantity from soil, crop and weather data.

24 Udyog Saarthi App - (Progressive Web-based Application) for Adults undergoing Job coaching for opportunities under 4% reservation in NIEPMD and other Institutions.

PS #205 SIH1333 PWA / MOBILE

Persons with disabilities in our country, despite the progress happening on accessibility, education, skills development, and employment, still face many challenges, especially in the employment phase of life that promotes inclusion in the family and society. Enhancing vocational training and employment prospects for people with disabilities is essential and there are efforts happening at the National level that has to end up in employment that fetches significant economic benefits and dignified life for them. As per the recent survey by the Ministry of Statistics report on Persons with disabilities, about 64% of persons with disabilities in India are not into employment. The vulnerable group of the population with disabilities under the D andamp; E categories of 4% reservation in Government and PSU Jobs (Persons with disabilities of type, Intellectual Disability, Autism Spectrum Disorder, Multiple Disabilities, Mental Illness, and Specific Learning Disabilities) are either not aware of or not getting notifications regarding the employment vacancies, queries, or other such information due to the unavailability of access to print media, and online services like employment news. They do require a curated and accessible content with adaptations to specifically inform them of the job and details in a simple language with requirement information to apply, know the coaching facilities available and get trained to apply, appear for competitive examinations, get qualified and place themselves to achieve the 1% reservation allocated to them. In many scenarios, the parents of such adults are also not with much literacy, awareness about such job opportunities through their adult children are eligible for them. This results in many of the jobs though published not being applied, for or the right candidates getting a placement as per the reservation and implementation of the Rights of Persons with Disabilities, Act 2016. Detailed Description: Accessibility and inclusion of Persons with disabilities (PwDs) in the workforce remains a challenging one on a global scale and with much more diversity in our Country's context. There have been ongoing attempts and measures to ensure the inclusion of PwDs in the workforce especially in the Government and PSUs level by the Department and other agencies working for them. This includes the Job Coach programme, an introduction to job support services, with support for application, training for competitive examination in an accessible manner, guidance to the PSUs and Department for accessible conduct of competitive examinations, counselling, and logistic support for candidates to write / attend the examination and interview thereafter. Finally, on placement, support for workplace adaptations or reasonable accommodation to sustain the job. The Department of Empowerment of Persons with Disabilities (DEPwD), National Institutions like NIEPMD, collaborating with Government and Non-Governmental Organizations (NGOs) support with such job coach programme and assist Persons with Disabilities (PwDs) especially D andamp; E category, the most vulnerable group with either cognitive, learning, severe mobility, behaviour management, socialisation, or multiple such difficulties. Expected Solution: The software application intends to Ãƒâ€šÃ‚Â· Create a learning profile of the candidates (The learning profile includes persons information with UDID, literacy, skills training, address, functional difficulties faced (as per Washington Group Extended Set Questions), assistive devices/solutions used or required, human assistance requirements, aspiration profile). Ãƒâ€šÃ‚Â· Provide information in an accessible manner to the candidates / parents for various opportunities happening as per 4% reservation in Government and PSUs. Ãƒâ€šÃ‚Â· Additionally, inform jobs as per equal opportunity policies happening in private sector. Ãƒâ€šÃ‚Â· Notification on important dates related to jobs application, examination, and interview. Ãƒâ€šÃ‚Â· Support for model question paper, practice, and mock tests before qualifying examinations. Ãƒâ€šÃ‚Â· Tracking for applied, examination attended, interview qualified and placement. Ãƒâ€šÃ‚Â· Workplace support requirements requests for facilitation. Ãƒâ€šÃ‚Â· Department of Adult Independent Living (DAIL), NIEPMD related other skills training programmes information notifications for getting enrolled, qualified with skills development. Source metadata Year: 2023 Domain / Technology Bucket: Smart Automation Category (SW/HW): Software Organisation: Ministry of Social Justice and Empowerment

> **Why it fits a 3-person team:** Udyog Saarthi employment app for people with disabilities — accessibility, job coaching and opportunities.

4. OTHER NOTABLE SOFTWARE STATEMENTS

Development of AI/ML based solution for detection of face-swap based deep fake videos PS #81 SIH1683 AI / ML

Detection of face-swap based deepfake videos.

Develop a ML Model based solution to refine CAPTCHA. PS #75 SIH1672 AI / ML

ML model based solution to refine CAPTCHA.

Fake Social Media Profile detection and reporting PS #229 SIH1364 AI / SECURITY

Fake social media profile detection and reporting.

Develop and deploy a Large Language Model (LLM) based tool for generating human like responses to natural language inputs for network not connected over internet PS #297 SIH1450 AI / SYSTEMS

LLM-based tool for human-like responses on offline, network-isolated machines.

Health Data Information & Management System Mobile Application (HDIMS) PS #54 SIH1626 MOBILE / HEALTH

Health Data Information & Management System (HDIMS) mobile application.

Learning App for Deaf And Mute and sign language-English/Gujarati converter PS #45 SIH1610 AI / ACCESSIBILITY

Learning app for the Deaf & Mute with a sign-language ↔ English/Gujarati converter.

Source: ADCMS “Select a Problem Statement — for Major Project” (368 of 371 statements available to select). Prepared for internal team review.